# Advanced Mathematical Economics

# First-Order Difference Equations General Solution

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# Outlines

- Difference Equations
- Types of Difference Equations
- First Order Linear Difference Equation
- Solution of First Order Linear Difference Equations
- Time Path
- Particular Function
- Complementary Function
- General Solution (if $a \neq -1$)
- Definite Solution (if $a \neq -1$)

# Outlines

- General Solution (if a=-1)
- Definite Solution (if a=-1)
- Summary of Lecture

# Difference Equations

- A difference equation is a mathematical equation that describes the relationship between values of a sequence or time series, where the value at one point in the sequence is related to the values at previous points through differences rather than derivatives.

# Types of Difference Equations

- There are various types of difference equations, but in this course, we study only:
- First-order difference equations.
- Higher-order difference equations.
- Non-linear difference equations

# First order Difference Equations

- The standard form of a 1st Order linear difference equation can be written as:

$$\boldsymbol{Y_{t+1} + aY_t = C \ldots \ldots \ldots (1)}$$

- Where:
- $Y_{t+1}$ is the value of the sequence at time *t+1*.
- $Y_t$ is the value of the sequence at time *t*.
- *a* is the coefficient multiplying the value $Y_t$.
- *c* is the constant term added to the equation.

# Non-Homogeneous Case (NHC), Where C≠0

- $\mathbf{Y_{t+1} + aY_t = C \ldots\ldots\ldots (2)}$
- **Where right hand side is non-zero**
- **Or Where C≠0**

# Homogeneous Case(HC), where C=0

- $\mathbf{Y_{t+1} + aY_t = 0 \ldots \ldots \ldots (3)}$
- **Where right hand side is zero**
- **Or C=0**

# Time Path

The time path is similar to what we have previously studied in differential equations, depending on the complementary function and particular function. The equation for the time path is given below:

- $$\boldsymbol{Y_t = Y_c + Y_p \ldots\ldots\ldots(4)}$$

- Note that differential equations depend on the derivatives of dependent and independent variables, whereas difference equations depend on the differences between current values, lead values, or lag values.

# Discrete time and difference Equations

- $\triangle \boldsymbol{Y_t} = \boldsymbol{Y_{t+1}} - \boldsymbol{Y_t} \ldots \ldots \ldots (\mathbf{5})$

- The equation tell us that;
- $\triangle Y_t$= change in $\mathrm{Y}_t$ depends on the difference between lead value $Y_{t+1}$ and $Y_t$ current value

# Derivation of Particular Function (Yp)

- Given NHC equation (2):
- $\mathbf{Y_{t+1}} + \mathbf{aY}_{\boldsymbol{t}} = \mathbf{C} \ldots\ldots\ldots(\mathbf{2})$
- Assume $\mathbf{Y_{t+1}} = \mathbf{Y}_{\boldsymbol{t}} = \mathbf{Y}_{\boldsymbol{p}}$
- we're essentially assuming that the sequence remains constant over time.
- In other words, $\mathrm{Y}_p$ represents a steady-state or equilibrium value of the sequence.

# Derivation of Particular Function (Yp)

- Substituting $\mathbf{Y_{t+1}} = \mathbf{Y}_{\boldsymbol{t}} = \mathbf{Y}_{\boldsymbol{p}}$into the difference equation (2)
- $\mathbf{Y_{t+1}} + \mathbf{aY}_{\boldsymbol{t}} = \mathbf{C}$, we get:
- $\mathbf{Y}_{\boldsymbol{p}} + \mathbf{aY}_{\boldsymbol{p}} = \mathbf{C}\ \ldots\ldots\ldots(\mathbf{6})$
- By taking $\mathbf{Y}_{\boldsymbol{p}}$ common
- $\mathbf{Y}_{\boldsymbol{p}}\,(\mathbf{1} + \mathbf{a}\quad) = \mathbf{C}.........(7)$
- Dividing $(\mathbf{1} + \mathbf{a}\quad)$**both sides**
- $\boldsymbol{Y_p} = \dfrac{\boldsymbol{C}}{(\mathbf{1}+\boldsymbol{a}\quad)}$ **.........(8)**

# Derivation of Complementary Function (Yc)

- For derivation $\mathrm{Y}_c$ we consider the Homogeneous Case from equation(3).
- $\mathbf{Y_{t+1}} + \mathbf{aY}_{\boldsymbol{t}} = \mathbf{0} \ldots\ldots\ldots(\mathbf{3})$
- Assume $\mathbf{Y}_{\boldsymbol{t}} = \mathbf{A}b^{t} \ldots\ldots\ldots(\mathbf{9})$
- $\boldsymbol{Y}_{\mathbf{t+1}} = \mathbf{A}\boldsymbol{b}^{\boldsymbol{t}+\mathbf{1}} \ldots\ldots\ldots(\mathbf{10})$
- Substitute (9) and (10) in equation (3), we get
- $\mathbf{A}\boldsymbol{b}^{\boldsymbol{t}+\mathbf{1}} + \boldsymbol{a}\mathbf{A}\boldsymbol{b}^{\boldsymbol{t}} = \mathbf{0} \ldots\ldots\ldots(\mathbf{11})$
- Or $\mathbf{A}\boldsymbol{b}^{\boldsymbol{t}} * \boldsymbol{b}^{\mathbf{1}}$**+** $\mathbf{aA}\boldsymbol{b}^{\boldsymbol{t}} = \mathbf{0}$**……….(12)**
- By taking common $\mathbf{A}b^{t}$
- $\mathbf{A}\boldsymbol{b}^{\boldsymbol{t}}$**(b+a)=0**

# Derivation of Complementary Function ($Y_t$ )

- Or
- **b=-a put in (9)**
- $\mathbf{Y_t = A}b^t \dots\dots\dots (\mathbf{9})$
- $\mathbf{Y_c = A}(-a)^t \dots\dots\dots (\mathbf{10})$

# Obtaining General Solution; if a≠0

- By substituting (10) and (8), in (4) we get
- $\boldsymbol{Y_t = Y_c + Y_p \ldots \ldots . (4)}$
- $\boldsymbol{Y_t = A(-a)^t + \frac{c}{(1+a\ \ )} \ldots . . (11)}$

# Obtaining Definite Solution; a≠-1

- For definite solution:
- Assume t=0 to find the value of Arbitrary constant A.
- Substitute **t=0 in General Solution, equation (11)**
- $Y_t = \mathbf{A}(-a)^t + \frac{c}{(1+a\ \ )}$ ..... **(11)**
- $Y_0 = \mathbf{A}(-a)^0 + \frac{c}{(1+a\ \ )}$
- $Y_0 = \mathbf{A}(1) + \frac{c}{(1+a\ \ )}$ ; **because** $(-a)^0$**=1**

# Definite Solution; a≠-1

- **Solving for A**
- **A=** $Y_0 - \frac{c}{(1+a\ )}$ **.........(12)**
- **Substitute the value of "A" in general solution (11), we get**
- $Y_t = \mathbf{A}(-a)^t + \frac{c}{(1+a\ )}$ ..... **(11)**
- $Y_t = \{Y_0 - \frac{c}{(1+a\ )}\}(-a)^t + \frac{c}{(1+a\ )}$ ..... **(13)**

# General Solution ; a= -1

- $Y_t = \mathbf{A} + Ct \ldots\ldots (\mathbf{14})$

# Definite Solution; if a= -1

- **Solving for A by assuming t=0**
- **By substituting t=0 in (14)**
- $Y_t = \mathbf{A} + Ct \ldots\ldots$ **(14)**
- $Y_0 = \mathbf{A} + C * 0$
- $Y_0 = \mathbf{A}$
- **Or,** $\mathbf{A} = Y_0 \ldots\ldots (15)$

# Definite Solution; if a= -1

- *By sub.* $\mathbf{A} = Y_0$ *in (14), we get*

- $Y_t = Y_0 + Ct \ldots\ldots (16)$

# Summary

- Satndard form of first order difference equation
- $\mathbf{Y_{t+1}} + \mathbf{a}\mathbf{Y}_{\boldsymbol{t}} = \mathbf{C}$
- Non-Homogeneous Case if C≠0
- $\mathbf{Y_{t+1}} + \mathbf{a}\mathbf{Y}_{\boldsymbol{t}} = \mathbf{C}$
- Homogeneous Case if C=0
- $\mathbf{Y_{t+1}} + \mathbf{a}\mathbf{Y}_{\boldsymbol{t}} = \mathbf{0}$

# Summary

- **The objective of difference equations is to find the time path:**
- $\boldsymbol{Y_t = Y_c + Y_p \ldots \ldots (4)}$
- **Particular Solution or equilibrium solution (Yp)**
- $\boldsymbol{Y_p = \frac{c}{(1+a\ \ )} \ldots\ldots\ldots(8)}$
- **Complementary faction (Yc)**
- $\mathbf{Y_c = A(-a)^t \ldots \ldots \ldots (10)}$

# Summary

- **Obtaining General Solution; if a≠0**
- $Y_t = \mathbf{A}(-a)^t + \frac{c}{(1+a\ \ )}$ ….. **(11)**
- **Definite Solution; a≠-1**
- **A**$= Y_0 - \frac{c}{(1+a\ \ )}$**………(12) if t=0**
- $Y_t = \{Y_0 - \frac{c}{(1+a\ \ )}\}(-a)^t + \frac{c}{(1+a\ \ )}$ ….. **(13)**

# Summary

- **General Solution ; a= -1**
- $\boldsymbol{Y_t = \mathrm{A} + Ct \ldots\ldots}$ **(14)**
- **Definite Solution; if a= -1**
- $\boldsymbol{\mathrm{A} = Y_0 \ldots\ldots (15) assuming\ t = 0}$
- $\boldsymbol{Y_t = Y_0 + Ct \ldots\ldots}$ **(16)**

- Thanks for watching!
- If you have any queries, please feel free to email or comment on the video.
- In the next video, we will solve numerical examples of difference equations from the book of Alpha C. Chiang for further clarification and comprehensive understanding.

- Thanks